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Branch:- AI&DS(A)

Batch:- A2

Experiment-2 : Data Ingestion and Cleaning

Aim:- To acquire data and prepare it for analysis using data ingestion and cleaning techniques

Theory :-

Data ingestion refers to the process of importing data from different sources (like CSV "files, databases, etc.) into a system for analysis. Cleaning the data is an important step because raw data often contains missing values, duplicates, and inconsistencies.

Key Steps in Data Cleaning :-

- Removing duplicate data
- Handling missing values
- Correcting inconsistent formats
- Changing data types
- Filtering required data

Tools Used :-

- Python
- Pandas Library

Programs :-

Basic Level (Weak Students)

1. Load CSV and Display First 5 Rows

```
from google.colab import files
uploaded = files.upload()
df = pd.read_csv("/content/data.csv")
import os
print(os.getcwd())
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head())
```

2. Identify Missing Values

```
print(df.isnull().sum())
```

3. Replace Missing Values with Zero

```
df.fillna(0, inplace=True)
print(df)
```

4. Rename Columns to Lowercase

```
df.columns = df.columns.str.lower()
print(df.columns)
```

5. Display Dataset Shape

```
print("Rows and Columns:", df.shape)
```

Moderate Level (Normal Students)

6. Remove Duplicate Rows

```
df = df.drop_duplicates()
print(df)
```

8. Change Data Types

```
df['age'] = df['age'].astype(int)
```

9. Filter Rows Based on Condition

```
filtered = df[df['age'] > 18]
print(filtered)
```

10. Save Cleaned Dataset

```
df.to_csv("cleaned_data.csv", index=False)
```

Advanced Level(Bright Students)

11. Compare Missing Value Strategies

```
mean_fill = df.fillna(df.mean(numeric_only=True))
median_fill = df.fillna(df.median(numeric_only=True))
print("Mean Strategy:\n", mean_fill.head())
print("Median Strategy:\n", median_fill.head())
```

12. Clean Inconsistent Categorical Values

```
df['gender'] = df['gender'].str.lower().replace({
    'male': 'M', 'm': 'M',
    'female': 'F', 'f': 'F'
})
```

13. Automate Cleaning Using Function

```
def clean_data(df):
    df = df.drop_duplicates()
    df.fillna(df.mean(numeric_only=True), inplace=True)
    df.columns = df.columns.str.lower()
    return df
df = clean_data(df)
```

14. Outlier Detection (IQR Method)

```
Q1 = df['age'].quantile(0.25)
Q3 = df['age'].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df['age'] < Q1 - 1.5*IQR) | (df['age'] > Q3 + 1.5*IQR)]
print(outliers)
```

15. Data Cleaning Report

```
print("Data Cleaning Report")
print("Shape:", df.shape)
print("Missing Values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
```

Conclusion

In this experiment, we learned how to ingest data from CSV files and perform essential data cleaning operations such as handling missing values, removing duplicates, and correcting inconsistencies. These steps improve data quality and make it suitable for accurate analysis and decision-making.